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Robo-advisors: A systematic literature review

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Robo-advisors: a systematic literature review

Abstract

Using a systematic literature review, our study analyzes articles on robo-advisors between the years 2017-2022. Our review identifies four relevant research streams: early classification of robo-advisors, behavioral topics, performance, and algorithm modelization. Finally, we propose relevant research questions for each stream, providing scholars with new research angles. Our considerations are also valuable for asset managers, banks, and other financial companies since the adoption of robo-advisors affects their business models through clients' preferences and cost structure.

1. Introduction

Robo-advisors are FinTech financial advisors that use artificial intelligence data processing (Bhatia et al., 2020; Tao et al., 2021) to provide investment advice with limited or no human support (Hodge et al., 2021; Tao et al., 2021). In addition to providing continuous financial advice (24 hours/7 days), Robo-advisors also produce more timely financial forecasts and reduce selection biases (Jung et al., 2018). This can lower the company cost of financial advice (Baker and Delleart, 2017) and increase profitability.

Robo-advisors are positioned to offer significant opportunities for financial companies by transforming the wealth management industry (Sironi, 2016). From the 297 billion USD in assets under management (AUM) in 2017, the worldwide robo-advisory market could presumably reach 2.76 trillion USD by the end of 2023 and 4.66 trillion USD by 2027.¹ The Covid-19 pandemic spurred the use of this technology (Isaia and Oggero, 2022), even though there are concerns about user satisfaction (Tiberius et al. 2022), their performance (Helms et al., 2022), and financial inclusion (Tan, 2020).

The rapid growth in the financial industry has also attracted growing scholarly interest in robo-advisors, and more research is still needed (Lucey et al., 2018) to provide managers and investors the necessary support to understand the robo-advisor phenomenon and how they deal with daily management decisions. Based on the aforementioned arguments, and using a systematic literature review (hereafter, SLR), this paper comprehensively outlines the current state of the art on robo-

¹ Robo-Advisors - Worldwide. (n.d.). Retrieved September 27, 2023, from <https://www.statista.com/outlook/dmo/fintech/digital-investment/robo-advisors/worldwide>.

advisors and seeks to answer the following research questions: what are the main research streams on robo-advisors? What are the possible new avenues for future research?

The remainder of the paper is structured as follows. Section 2 describes the research design and Section 3 analyzes and discusses the main findings and proposes future research questions. Finally, Section 4 concludes.

2. Research design

To depict the current state of the literature on robo-advisors, we employ a systematic literature review since a SLR is particularly suitable when the topic is characterized by heterogeneity and because it guarantees a replicable and reliable understanding of current knowledge (Tranfield et al., 2003; Thorpe et al., 2005).

We use two databases to identify scientific articles on robo-advisors: Scopus and Web of Science. Thus, we search the keywords “Robo-advisor” or “Roboadvisor” and their corresponding plural terms, obtaining 199 total articles in English (103 in Scopus and 96 in Web of Science).² Finally, we remove duplications (77 articles) and papers outside the main topic of robo-advisors (19 articles). Thus, the final sample of eligible articles consists of 103 research papers (Table 1).

<Table 1 approximately here>

3. Findings and Discussion

3.1 Overview of publications

Figure 1 depicts an overview of the growth in publications on robo-advisors, suggesting that, although it is quite a recent phenomenon, it has captured the interest of researchers over the last few years. The papers reviewed were published in more than 80 different scientific Journals.

<Figure 1 approximately here>

3.2 Thematic analysis

We identify four research clusters on robo-advisors: (1) early attempts to classify robo-advisors; (2) behavioral topics; (3) performance; (4) algorithm modelization.

3.2.1 Early attempts to classify robo-advisors

The first strand of literature concerns an early attempt to classify robo-advisors. Based on the original work by Deloitte (2016), the first classification moves from the idea that the wealth management industry may employ robo-advisors with a different degree of intensity in active and passive

² We downloaded the selected papers in October 2022.

management strategies (Tokic, 2018). In passive investment strategies, robo-advisors assist investors in asset allocation (robo-advisor 1.0), risk management (robo-advisor 2.0), and proposing portfolio rebalancing (robo-advisor 3.0). Concerning active management strategies, investors benefit from fully automated investments based on artificial intelligence algorithms as well as proposing automatic asset shifts in clients' accounts (robo-advisor 4.0). An example of a robo-advisor 4.0 is BlackRock (Tokic, 2018).

A second classification of robo-advisors comes from the analysis of the different roles played by investors and the robo-advisor owners (Garvía, 2018). Specifically, the author distinguishes four types of robo-advisors: i) stand-alone robo-advisors, ii) segregated robo-advisors, iii) integrated robo-advisor, and iv), robo-for advice. A stand-alone robo-advisor is independent of other financial institutions, and thus robo-advisors manage the overall investment process internally. Segregated and integrated robo-advisors differ in their level of integration with other financial institutions, thus, although they are both linked to a financial institution. Segregated robo-advisors are part of a financial holding while integrated robo-advisors are included in the business model of the financial institution, such as banks. Finally, a robot-for advice identifies a tool that may be used by a set of financial institutions for a fee.

Beyond identifying independent robo-advisors and robo-advisors as part of an investment holding, the third classification identifies the D2C model, the B2B model, and the hybrid firms. The D2C model identifies “online platforms that provide automated algorithms based on portfolio management without human intervention”, while the B2B model identifies a robo-advisor that digitally supports a human advisor. Finally, hybrid firms provide “personalized services and actively managed portfolios blended with computerized portfolio recommendations” (Phoon and Koh, 2017, p. 9).

To summarize, the classifications of robo-advisors and highlight the flexibility of this FinTech phenomenon: robo-advisors help active and passive management strategies, represent either a business line or the overall business of a firm, and support a human financial advisor or work alone as an automated financial assistant. Thus, this flexibility provides multiple options for investors and financial institutions.

3.2.2 Behavioural

Another strand of literature tackles the behavioral aspects driving the customers' intention to rely on robo-advisor services, the factors affecting the investment procedure, and the decision to continue using robo-advisors (for a synthesis, see Table 2).

Concerning customers' intentions to use robo-advisors, we identify five specific grounds: investors' personal attributes, characteristics related to the investment style, features of human advisors that

indirectly link to the use of their robo-advisor peers, features of robo-advisors, and finally, risks connected with robo-advisors.

Turning to the investors' demographic profiles explaining the usage of robo-advisory services, the literature identifies the main characteristics of their current and potential users. First, and unsurprisingly, older investors are less likely to use robo-advisors (Brenner and Meyll, 2020; Fan and Swarn, 2020; Piehlmaier, 2022) and robo-advisors sustainability-oriented (Au et al., 2021). Second, except for Brenner and Meyll (2020), female and male customers are equally likely to use robo-advisors (Flavián et al., 2022, Figà-Talamanca et al., 2022). Then, the personality traits of clients, such as extroversion, optimism, risk-attitude (Oehler et al., 2022), and the fear of being a victim of financial fraud (Brenner and Meyll, 2020) positively affect customer likelihood of employing robo-advisors for their investments. Additionally, individuals' financial literacy (Isaia and Oggero, 2022) and overconfidence in their financial expertise (Piehlmaier, 2022) act as positive drivers for robo-advisor use, such as service awareness (Flavián et al., 2022). Also, the user's relationship with technology matters. On the one hand, technological optimism, innovativeness, and interestingly, discomfort urge people to use robo-advisors. On the other hand, technology insecurity (Flavián et al., 2022) and algorithm aversion (Filiz et al., 2022) dampen the use of robo-advisors. Finally, the customers' income (Piehlmaier, 2022) is not a distinctive feature affecting the use of robo-advisors. The relevant features of the client investment style that seem to encourage the use of robo-advisors include: having a limited financial portfolio or an innate preference for standardized financial products, such as exchange-traded funds (Brenner and Meyll, 2020), and using mobile banking (Wang et al., 2021). Finally, Nourallah et al. (2022), confirm that the rational decision style³ has a positive impact on the use of robo-advisors. Indeed, potential rational users search for information about robo-advisors, build their trust in the instrument, and finally decide to use a robo-advisor.

Scholars also underscore that robo-advisor-specific characteristics affect client probability of usage. Ease of usage is one of the most important key factors for robo-advisor use (Atwal and Bryson, 2021; Figà-Talamanca et al., 2022; Seiler and Fanenbruck, 2021; Yeh et al., 2022). Figà-Talamanca et al. (2022), however, highlight intergenerational differences in ease of use and usefulness, with the ease of use mostly relevant for Generation Z and Y, and usefulness for Generation X (Figà-Talamanca et al., 2022). Performance (Ge et al., 2021; Yeh et al., 2022) and privacy (Seiler and Fanenbruck, 2021) positively affect the investors' choice of using robo-advisors. In this context, information source about the robo-advisors matters. For instance, an in-depth description of robo-advisor algorithms

³ Nourallah et al. (2022) explain that "A rational decision style refers to individuals who conduct a comprehensive search for information and a systematic assessment of all potential choices. In contrast, an intuitive decision style refers to individuals who use a quick decision-making process that relies on hunches and feelings (Hamilton et al., 2017)".

(Litterscheidt and Streich, 2020) or social media coverage (Nourallah et al. 2022; Belanche et al., 2019) play a positive role, as well as users' acquaintances (Belanche et al., 2019).

The literature also highlights the potential risks associated with investments based on robo-advisors that might affect their usage. Nourallah et al. (2022) classify these risks as financial, performance, security and privacy, and social. Yet, the latter does not influence the decision to use a robo-advisor (Nourallah et al., 2022), while Atwal and Bryson (2021) find a negative influence of perceived risks⁴ in choosing to use a robo-advisor.

The decision to accept suggestions from a robo-advisor (Investment-phase, Table 2) is under-explored in the literature. One exception is the study by Hildebrand and Bergner (2020), who argue that if the recommendations are made by a Conversational robo-advisor with “an advisory interfaces that possess a dialogue-based process of financial advisory” (p. 559), investors are much more likely to accept the recommendation.

Finally, regarding the decision of confirming trust in a robo-advisor or using a human advisor, Maier et al. (2022) find that misalignments in performance between robo-advisors and human financial advisors produce mistrust and avoidance of robo-advisors and preference shifts to human advisors for portfolio allocation. By contrast, network size and complementarity, gamification, information, system, and service quality may positively impact the decision to continue using a robo-advisor (Cheng, 2022).

<Table 2 approximately here>

3.2.3 Performance

The third stream of the literature covers measuring the risk-adjusted performance of the robo-advisor against market benchmarks and human advisors and provides evidence of benefits for a set of investors and asset managers.

Regarding robo-advisor performance against market indexes, the literature suggests that robo-advisors do not outperform market indexes either in a static setup or in a dynamic setup when the empirical framework allows for portfolio rebalancing. This finding is corroborated by Shanmuganathan (2020), comparing four robo-advisors - Betterment, Wealthfront, FutureAdvisor, and Charles Schwab - and by Puhle (2019), considering five German robo-advisors - Easyfolio, Fintego, Quirion, SutorBank, Vaamo. Horn and Oehler (2020) find that even automated portfolio

⁴ Risk is defined to include various dimensions: “(1) performance risk, (2) financial risk, (3) time risk, (4) psychological risk, (5) social risk, (6) privacy risk, and (7) security risk” (Akturan & Tezcan, 2012).

rebalancing does not yield a better performance than static investment strategies, such as a buy-and-hold strategy.

Moving onto the sub-strand of the literature comparing the performance of robo-advisors and human advisors, Harrison and Samaddar (2020) highlight that robo-advisors do not outpace human advisors in a simulated context, while Tao et al. (2021) find some evidence that robo-advisors outperform human advisors. Helms et al. (2022), comparing fifteen robo-advisors from Germany, the United Kingdom, and the United States of America,⁵ find that robo-advisors outperform human advisors. However, they also identify the source of divergence in their performance as mainly related to the business model and the target client profiles. Yet, they also show that managerial fees and rebalancing styles have only a marginal impact on investment performance. In fact, a robo-advisor's performance varies as a function of the investors' degree of diversification.

Another perspective of analysis is whether robo-advisors provide financial performance benefits for investors with under-diversified investment portfolios (D'Acunto et al., 2019) and for low-income and/or low-education individuals (D'Hondt et al., 2020). Empirical results confirm these hypotheses. Finally, an under-investigated topic is whether robo-advisors positively affect the performance of asset managers. In this perspective, Deng et al. (2021) document that a robo-advisor is highly beneficial for small and medium-sized banks. Table 3 summarizes robo-advisor performance.

<Table 3 approximately here>

3.2.4 Modelization

The final stream of literature concerns the algorithm modelization driving the functionality of the robo-advisor. Algorithms can include traditional theories and assets, or alternative ones, such as crypto assets (e.g., Babaei et al., 2022; Giudici et al., 2020).

Although Modern Portfolio Theory represents the most common basis for robo-advisor algorithms (Beketov et al., 2018), some scholars propose more sophisticated models. For example, they can be based on the user's spending goals according to their risk tolerance (for example, Kim et al., 2020), while Alsabab et al. (2021) and Dong et al. (2022) propose robo-advisor investment algorithms based on investor preferences related to their previous investment choices to understand their financial profiles rather than relying on questionnaires. Similarly, Capponi et al. (2022) develop a robo-advisor algorithm that timely rebalances the financial portfolio to allow for changes in investor preferences.

3.3 Discussion and Future Research

⁵ Germany: Scalable, Cominvest, Liquid, Quirion, Growney; The United Kingdom: Nutmeg, Moneyfarm, Wealthsimple, Wealthify, Scalable; The United States: Schwab Intelligent, Betterment, Weath Front, SigFig, Axos Invest.

Robo-advisors represent a growing research field. We identified four main streams. First, scholars classified stereotypes of robo-advisors (e.g., Garvia, 2018; Phoon e Koh, 2017; Tokic, 2018), emphasizing the managerial and technical aspects of robo-advisors, with a brief focus on their implications for investors and investors' protection. Analyzing further regulatory stances of robo-advisors might open a new research stream for this literature.

Second, a large number of studies dissect the behavioral aspects explaining the customers' decision to employ robo-advisor services for investment purposes as well as factors affecting the investment procedure itself, and the decision to continue using robo-advisors for further investment decisions. In our view, two aspects appear to be unexplored. On the one hand, it is still unclear which factors increase/decrease the investors' satisfaction, engagement, and trust in comparison with traditional advisors. Gathering more evidence through (international) surveys could furnish valuable information on this specific facet. On the other hand, given the rising importance of sustainable investing and changes in investors' preferences related to sustainability issues (e.g., Bassen et al., 2023), the understanding of more sustainable investors' preferences toward robo-advisors would be useful both for scholars and practitioners (e.g., asset managers and banks) strengthening this business line.

The third strand of literature benchmarks robo-advisors' profitability with indexes and human advisors. Additionally, scholars consider the wealth effects for low-income/low-education investors with under-diversified portfolios and asset managers. Some lines of research are currently lacking and may benefit from investigation. **For instance, it would be interesting to have a deeper understanding of the effects of robo-advisor activity compared with the activity of innovative chatbots for the portfolio allocation and portfolio rebalancing, as more recently suggested by Oehler and Horn (2024).** This is also relevant for investors to understand the related implications for investors' protection. Further, it would be interesting to explore the final effect of robo-advisor adoption on asset managers' risk-adjusted performance. Current evidence is limited to a set of banks. This facet is also important from a regulatory standpoint since robo-advisor activity affects the financial intermediary's efficiency and their financial stability.

The stream related to algorithms and modelization focuses on traditional theories. For instance, future research should be devoted to integrating more aspects of behavioral finance into algorithmic models to better predict investors' needs and tailor their investment solutions.

Finally, beyond these four strands highlighted by the prior literature, we propose a further field of investigation concerning the socioeconomic impact of robo-advisors, including their role in boosting financial inclusion and their effects on traditional employment in the financial sector.

Table 4 describes new avenues for future research.

<Table 4 approximately here>

4. Conclusion

Our systematic review is valuable for scholars because it sheds light on the main streams of research on robo-advisors and provides a novel set of research avenues. The review is also a useful toolkit for asset managers, banks, and other financial companies interested in implementing or expanding robo-advisor operations. The review summarizes some relevant aspects for financial management (archetypes of robo-advisors, customers' characteristics and preferences, performance against benchmarks, human advisors and overall impact on asset managers' efficiency) and clarifies what is presently unclear (e.g., potential impacts on financial firms' employees). Finally, the review provides important supervisory and regulatory insights because of the potential impact of robo-advisors on the bank business model and efficiency, which is still currently unexplored in the literature.

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Table 1 – Search process and results

Database	Number of journal articles
Papers in Scopus	103
Papers in Web of Science	96
Database search total	199
Duplications	77
Exclusion of off topic papers	19
Selected publications	103

Table 2 – Behavioural factors

Category	Sub-category	Features	Authors, Years	Relationship with Robo-advisors
Intention of using a robo-advisor or usage of robo-advisors	Personal features of potential user or user	Age	Brenner and Meyll (2020)	Positive (young)
			Piehlmaier (2022)	Negative (old) Not significant (age 35-40 and digital natives)
			Flavián et al. (2021)	Not significant
			Fan and Swarn (2020)	Negative (more than 65 years) Positive (more than 65 years under some conditions)
			Au et al. (2021) *	Positive (young)
		Gender	Brenner and Meyll (2020)	Positive (male)
			Figà-Talamanca et al. (2022)	Not significant
			Flavián et al. (2021)	Not significant
		Afraid to be a victim of fraud	Brenner and Meyll (2020)	Positive
		Extroversion	Oehler et al. (2022)	Positive
		Optimism	Oehler et al. (2022)	Positive
		Risk-taking behavior	Oehler et al. (2022)	Positive
			Fan and Swarn (2020)	Positive
		Financial literacy	Isaia and Oggero (2022)	Positive
			Fan and Swarn (2020)	Positive
		Overconfidence in their financial knowledge	Piehlmaier (2022)	Positive
		Experience	Au et al. (2021) *	Positive
		Algorithm aversion	Filiz et al. (2022)	Negative
		Income	Piehlmaier (2022)	Not significant
		Customers' technology optimism	Flavián et al. (2021)	Positive
		Customers' technology innovativeness	Flavián et al. (2021)	Not significant
		Customers' technology discomfort	Flavián et al. (2021)	Positive
		Customers' technology insecurity	Flavián et al. (2021)	Negative
		Service awareness	Flavián et al. (2021)	
		Sustainable consumer	Bruner and Laubach (2022)	Positive
	Features of Human advisors	High expert human advisors in a high-risk domain	Zhang et al. (2021)	Negative
	Investment attitude/style	Financial portfolio	Brenner and Meyll (2020)	Positive (limited)
			Fan and Swarn (2020)	Positive (high amount)
			Northey et al. (2022)	Negative (high amount)
		Standardized financial products	Brenner and Meyll (2020)	
		Intuitive decision style	Nourallah (2022)	Positive
		Rationality	Nourallah (2022)	Positive
		Mobile banking user	Wang et al. (2021)	Positive
		Defaulted investor	Ge et al. (2021)	Negative
		Performance	Yeh et al. (2021)	Positive

	Features of robo-advisors and public information		Ge et al. (2021)	Positive
		Usefulness (or perceived usefulness)	Figà-Talamanca et al. (2022)	Positive (generation X)
		Ease of use (or perceived ease of use)	Figà-Talamanca et al. (2022)	Positive (generation z and y)
			Seiler and Fanenbruck (2021)	Positive
			Atwal and Bryson (2021)	Positive
			Yeh et al. (2021)	Positive
		Privacy	Seiler and Fanenbruck (2021)	Positive
		Named robo-advisor	Hodge et al. (2021)	Negative Positive (if tasks are simple)
		Expertise	Lovrenco et al. (2020)	Positive
		Amount of public info	Nourallah et al. (2022)	Positive
		Social media information	Nourallah et al. (2022)	Positive
		Detailed info on algorithm investment style	Litterscheidt and Streich (2020)	Positive
		Social influence / subjective norms	Belanche et al. (2019)	Positive
	Investment Risks	Financial risks	Nourallah et al. (2022)	Not significant
		Performance risks	Nourallah et al. (2022)	Not significant
		Security and privacy	Nourallah et al. (2022)	Not significant
		Social risks	Nourallah et al. (2022)	Not significant
		Perceived risks: (1) performance risk, (2) financial risk, (3) time risk, (4) psychological risk, (5) social risk, (6) privacy risk, and (7) security risk.	Atwal and Bryson (2021)	Negative
Investment phase	Features of robo-advisor	Conversational robo-advisors	Hildebrand and Bergner (2020)	Positive
Investment replication	Period of investment/replicated investment choice	Network size and complementarity	Cheng (2022)	Positive
		Gamification	Cheng (2022)	Positive
		Information, system, and service quality	Cheng (2022)	Positive

Table 3 Robo-advisors performance

Category	Authors, Years	Outperformance/underperformance/neutral
Performance against market indexes	Shanmuganathan (2020)	Neutral
	Puhle (2019),	Neutral
	Horn and Oehler (2020)	Neutral
Performance against human advisors	Harrison and Samaddar (2020)	Neutral
	Helms et al. (2022)	Outperformance
	Tao et al. (2021)	Outperformance
Investors performance	D'Acunto et al., 2019	Outperformance for low diversified portfolio
	D'Hondt et al. (2020)	Outperformance for low-income and low educated investors
Asset managers performance	Deng et al. (2021)	Outperformance for small and medium banks

Table 4 Future research questions

Cluster	Future research questions
Attempt to classify robo-advisors	- Does regulation play a role in classifying robo-advisors?
Behavioral topics	- Which factors increase/decrease the investors' satisfaction, engagement, and trust in comparison with traditional advisors? - Are sustainable-oriented investors more or less likely to use robo-advisors?
Performance	- Do robo-advisors are beneficial for portfolio allocation? - Do a wide-spread adoption of robo-advisors affect financial efficiency and stability? - What is the effect of robo-advisor adoption on the risk-adjusted performance of financial intermediaries? - Do regulatory discrepancies among countries impact robo-advisors performance and level of services? - Do robo-advisors impair investors' protection?
Algorithms and modeling	- How does the investors' customization affect robo-advisors algorithms?
Socioeconomic impact	- Do robo-advisors boost financial inclusion? - Do robo-advisors affect asset managers employee productivity?

Figure 1 Publications per year

